

Consumer Banking Complaints Analysis

Project Overview

This project utilizes SQL and Power BI to analyze consumer banking complaints to identify complaint trends, product-level patterns, common issues, and company response outcomes.

The analysis uses Consumer Financial Protection Bureau (CFPB) complaint data covering May-July 2026. SQL was used to clean, aggregate, and analyze the data, while Power BI was used to create an interactive dashboard to communicate key findings through visualizations.

Business Questions

This analysis aims to answer the following questions:

- Which banking products get the most consumer complaints?
- What are the most common issues in these complaints?
- How did the number of complaints change each month?
- Which companies got the most complaints?
- How do companies respond to consumer complaints?
- What percentage of complaints receive timely responses?
- Which products have the highest rates of monetary relief?

Key Findings

- The data set contains 55,271 consumer complaints
- Credit card complaints were the most common, with 21, 203 complaints
- Checking or savings accounts followed closely with 20,738 complaints
- May had the highest complaint volume at 19,802 complaints
- 67.25% of complaints were closed with an explanation
- 99.07% of complaints received a timely response
- Credit card complaints had the highest monetary relief by products with 2677 monetary relief complaints

Tools and Technologies

- SQL: Data Analysis and aggregation
- DB Browser for SQLite: SQL development and querying
- Power BI: Data visualization and dashboard development
- DAX: KPI calculations in Power BI
- GitHub: Project Documentation and portfolio presentation

Data

The analysis uses consumer complaint data from the Consumer Financial Protection Bureau (CFPB). The data set includes information such as:

- Product
- Sub-product
- Issue

- Sub-issue
- Company
- Company response to consumer
- Timely response
- State
- Complaint submission method
- Complaint date

SQL Analysis

SQL was used to analyze complaint volume, trends, company activity, and response outcomes.

Key Analyses included:

- Total complaint volume
- Complaints by product
- Top complaint issues
- Top companies by complaint volume
- Monthly complaint trends
- Company response outcomes

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51 -- 6. COMPANY RESPONSE TO CONSUMER
52 SELECT
53     "Company response to consumer",
54     COUNT(*) AS total_complaints,
55     ROUND(
56         COUNT(*) * 100.0 /
57         (SELECT COUNT(*) FROM CFPB_Complaints_May_July_2026),
58         2
59     ) AS percentage
60 FROM CFPB_Complaints_May_July_2026
61 GROUP BY "Company response to consumer"
62 ORDER BY total_complaints DESC;
63

```

	Company response to consumer	total_complaints	percentage
1	Closed with explanation	37169	67.25
2	In progress	9632	17.43
3	Closed with monetary relief	4997	9.04
4	Closed with non-monetary relief	3212	5.81
5	Untimely response	261	0.47

- Timely response rate
- Top sub-issues

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87
88 -- 8. TOP 10 SUB-ISSUES
89 SELECT
90     "Sub-issue",
91     COUNT(*) AS total_complaints
92 FROM CFPB_Complaints_May_July_2026
93 WHERE "Sub-issue" IS NOT NULL
94     AND "Sub-issue" <> ''
95 GROUP BY "Sub-issue"
96 ORDER BY total_complaints DESC
97 LIMIT 10;
98
99
100 -- 9. RESPONSE OUTCOMES BY PRODUCT
101 SELECT
102     Product

```

	Sub-issue	total_complaints
1	Credit card company isn't resolving ...	5609
2	Deposits and withdrawals	3914
3	Problem using a debit or ATM card	3383
4	Transaction was not authorized	2450
5	Company closed your account	2422
6	Card opened without my consent or ...	1663
7	Funds not handled or disbursed as ...	1416
8	Problem with fees	1374
9	Card was charged for something you ...	1360
10	Escrow, taxes, or insurance	1305

- Monetary relief by product

```

95 -- 8. Monetary Outcomes by Product
96 SELECT
97     Product,
98     COUNT(*) AS total_complaints,
99     SUM(
100         CASE
101             WHEN "Company response to consumer" = 'Closed with monetary relief'
102             THEN 1
103             ELSE 0
104         ) AS monetary_relief_complaints,
105     SUM(
106         CASE
107             WHEN "Company response to consumer" = 'Closed with monetary relief'
108             THEN 1
109             ELSE 0
110         ) * 100.0 / COUNT(*),
111     ) AS monetary_relief_rate
112 FROM CFPB_Complaints_May_July_2026
113 GROUP BY Product
114 ORDER BY monetary_relief_rate DESC;

```

	Product	total_complaints	monetary_relief_complaints	monetary_relief_rate
1	Credit card	21203	2677	12.63
2	Checking or savings account	20738	2121	10.23
3	Vehicle loan or lease	5566	101	1.81
4	Mortgage	7764	98	1.26

- Timely response rate by product

```

95 -- 7. TIMELY RESPONSE RATE
96 SELECT
97     SUM(
98         CASE
99             WHEN "Timely response?" = 'Yes' THEN 1
100             ELSE 0
101         ) AS timely_responses,
102     COUNT(*) AS total_complaints,
103     SUM(
104         CASE
105             WHEN "Timely response?" = 'Yes' THEN 1
106             ELSE 0
107         ) * 100.0 / COUNT(*),
108     ) AS timely_response_rate
109 FROM CFPB_Complaints_May_July_2026
110 -- 8. TOP 10 SUB-ISSUES

```

	timely_responses	total_complaints	timely_response_rate
1	54756	55271	99.07

Power BI Dashboard

The Power BI dashboard transforms the SQL analysis into an interactive report designed to provide an executive overview and deeper analysis of company performance.

Executive Dashboard

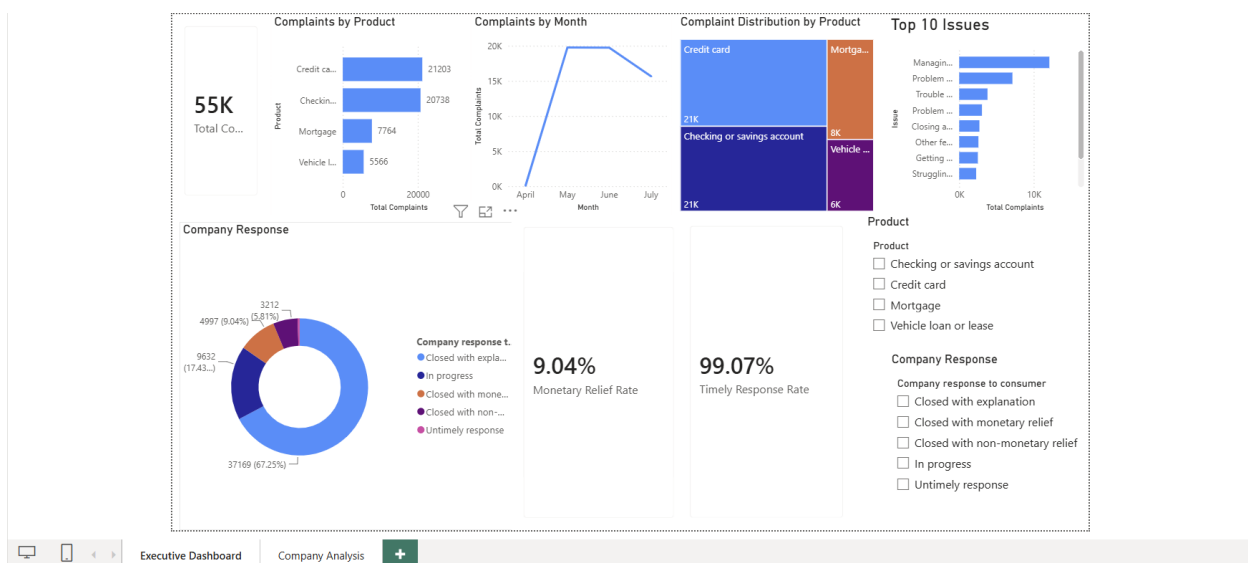
The Executive Dashboard provides an overview of consumer complaint activity.

Key metrics include:

- Total Complaints
- Total Companies
- Timely Response Rate
- Closed with Explanation Rate

Visualizations include:

- Complaints by Month
- Complaints by Product
- Top 10 Issues

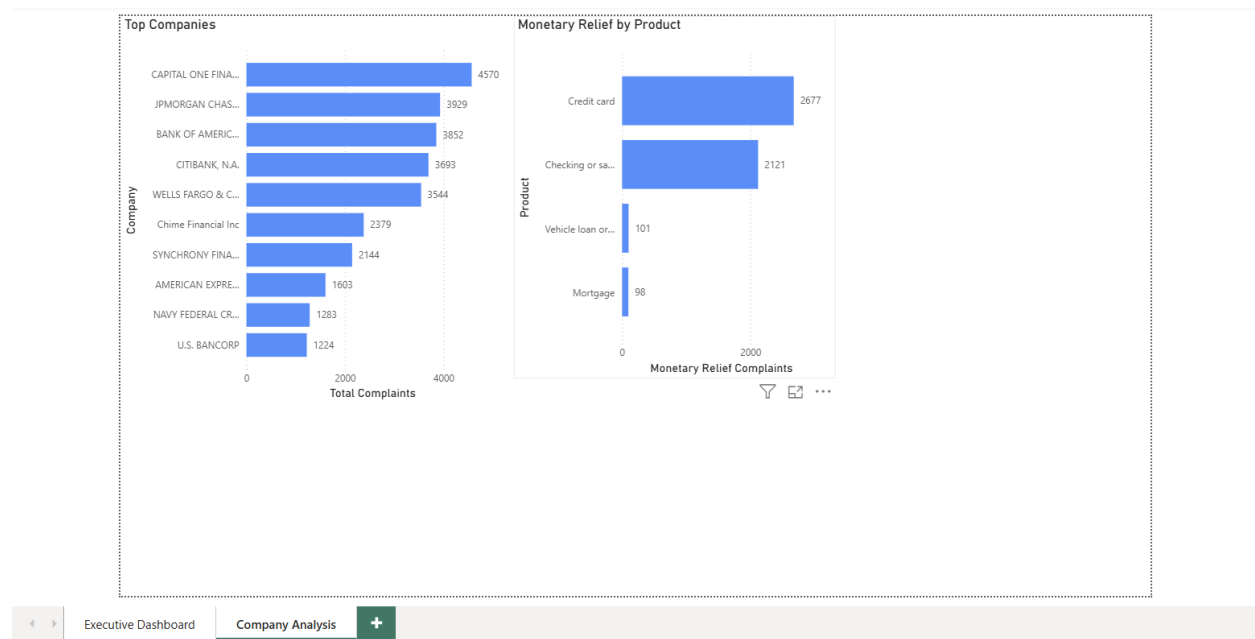


Company Analysis

The Company Analysis page focuses on company response patterns and product-level outcomes.

Visualizations include:

- Top Companies
- Monetary Relief by Product



Business Insights

The analysis shows that consumer complaints mainly focused on credit cards and checking or savings account products. Together, these products made up most of the complaints from May to July 2026. This suggests that they are the biggest source of consumer issues and may need more attention from financial institutions.

Complaint volume was highest in May, with 19,802 complaints, before declining in June and July. This indicates high level of fluctuation in complaint activity during the analysis period.

The analysis also shows that 67.25% of complaints were closed with an explanation, while 99.07% received a timely response. This suggests that companies generally respond within the expected timeframe, although timely responses do not indicate that the consumer issue was resolved.

At the issue level, recurring issues accounted for substantial complaint volume. The most common sub-issue was credit card companies not resolving consumer complaints, followed by deposits and withdrawals and problems using debit or ATM cards. This suggests that recurring service and transaction-related issues may represent important opportunities for financial institutions to improve the overall consumer experience.

Monetary relief also varied by product. Credit card complaints had the highest monetary-relief rate at 12.63%, followed by checking or savings accounts at 10.23%, while vehicle loans and mortgages had substantially lower rates. This indicates that complaint volume alone does not fully capture the financial impact associated with consumer complaints.

Business Implications

- Prioritize high-volume products: Credit cards and checking or savings accounts should receive more attention, as they receive more complaints
- Focus on recurring issues: Repeated complaints, which include unresolved credit card issues, deposits and withdrawals, and debit/ATM cards indicate opportunities for improvement
- High timely response: A high timely response rate (99.07%) indicates strong responsiveness, but organizations must decide whether complaints are being efficiently resolved

- Monitor financial impact: Products with higher monetary relief rates may require more investigation, as complaints involving these products are usually associated with financial remediation
- Monitoring complaint data: Complaint trends could be used to help identify areas where customer service regulations may need improvement.

Project Files

CFPB Complaints May-July 2026.csv (Source dataset)

CFPB Banking Complaints Analysis.sql (SQL analysis queries)

Consumer Banking Complaints.db (SQLite database containing the imported dataset)

Consumer Banking Complaints.sqbpro (DB Browser for SQLite project file)

Power BI Dashboard.pbix (Interactive Power BI report)

Company Response to Consumer.png (SQL analysis result)

Monetary Relief by Product.png (SQL analysis result)

Timely Response Rate.png (SQL analysis result)

Top 10 Sub-Issues.png (SQL analysis result)